YAKEEN

NEET 2.0 2027

Physics | Lecture - 05

Motion in a Straight Line

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Todays goal

Questions on velocity & acceleration

H/w Question ko kitne log read kar rahe hai"

H/w goal

(1) differentiation & Integration ka revision

(2) Sare assignment of 1 & 2

(3) Check last Page of P.P.T.

Uniform Velocity (Constant velocity) → Bhot udas hai motion

↳ No change in speed (How fast) and direction.

↳ Path must be straight line, No U-turn

↳ speed = cost^n acceleration = 0 = $\frac{\Delta v}{\Delta t}$

↳ |Avg velocity| = Inst. Velocity ✓ = Avg speed

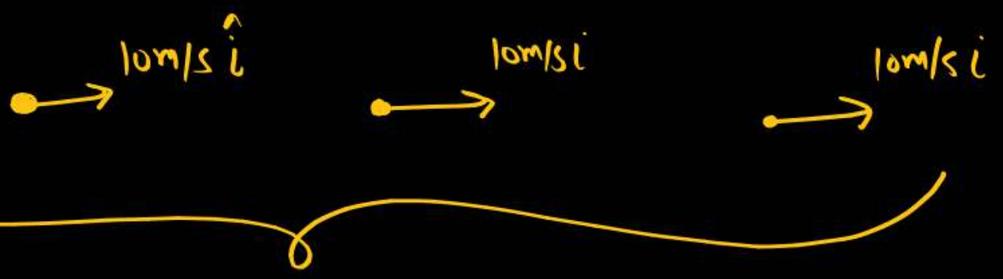
↳ **Uniform motion** ⇄ Uniform Velocity

Velocity = Speed × dir^n

↓ ↓ ↓

Cost^n Cost^n Cost^n (fixed)

only const when speed and dir^n both are const



Uniform speed (constant speed)

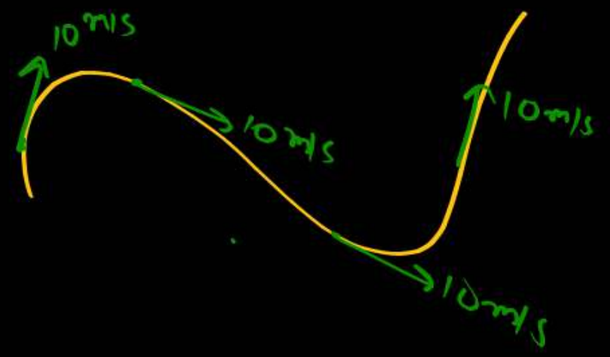
↳ How fast is Constant, but direction is not Known, Path not Known

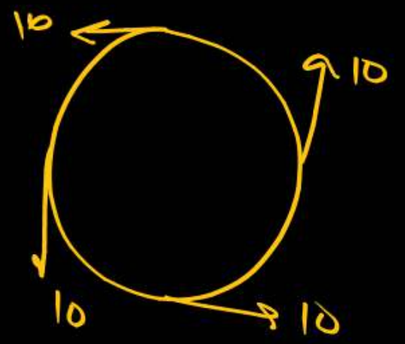
Motion may be 1-D or 2-D.

↳ velocity may or may not be uniform (constn)



↳ Uniform motion ✗





Instantaneous speed = How fast

- Scalar ✓
- Rate of change in distance w.r.t time
- $S = \left(\frac{dx}{dt}\right)$
  - slope of distance/time graph

{Magnitude of ~~Inst.~~ Velocity}

Inst. Velocity = How fast / where (dirn)

= speed x dirn

- Vector ✓
- * Rate of change in displacement ✓

$$\vec{v} = \frac{d\vec{x}}{dt}$$

- * slope of disp^m/time graph ✓
- * direction → along motion or tangent to path

MR* Box

use differentiation for gst. speed and gnst. velocity Not for Avg. velocity & Avg. speed.

(34) Diretion of motion changes at instant (time) ??

(U-turn kha liya hai)

(a) at A only

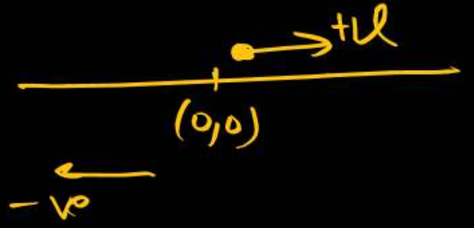
(b) at B only

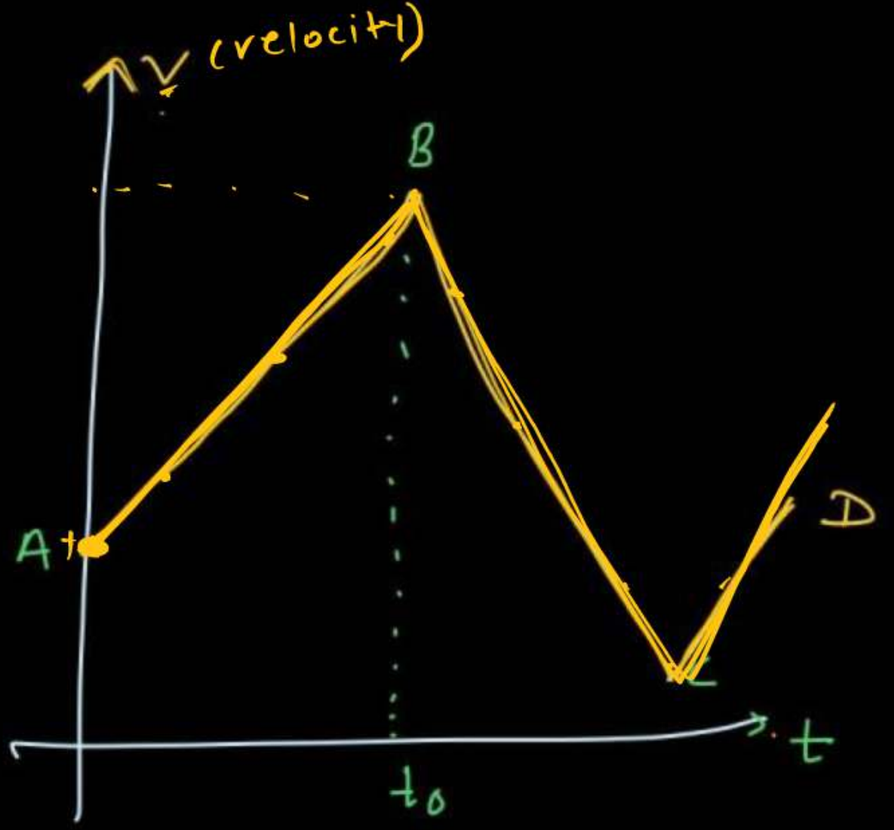
(c) at C only

~~(d) at B & C (-1)~~

(e) ✓ Non of these

41 % (+1) Pen





(35)

If inst. speed is Constant then Avg speed will be ??

→ Cosn

'Avg speed = inst speed

(36)

If inst velocity is Constant then Avg velocity will be ??

→ Avg velocity = ist. velocity

(37) Object is moving with constant speed then what about velocity ??

speed = cost^{n}.



Ans → velocity may or may not cost^{n}.

(38) Object is moving with constant velocity then what about speed

↓

Speed must be cost^{n}

(39) If object moving with **variable speed** then what about velocity

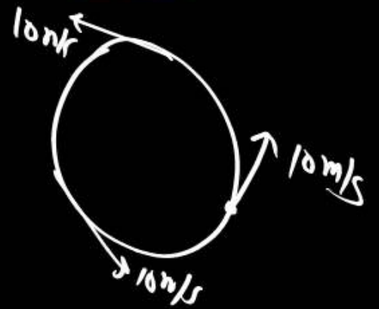
→ speed is changing

Velocity = speed × dirⁿ

↓

Ans → must be variable ✓

(40) If **velocity is variable** then **speed** of object will be ??



Velocity = Speed × dirⁿ = Variable

Ans

May or may not be change (Variable)

(41) gf Position $x = \alpha t^2 + \beta t - \gamma$ α, β, γ are Cosⁿ

(i) find itial velocity & position

(ii) find velocit at $t = 2$ sec

$x = \alpha t^2 + \beta t - \gamma$

$\left(\frac{dx}{dt}\right) = V = \alpha[2t] + \beta(1) - 0$

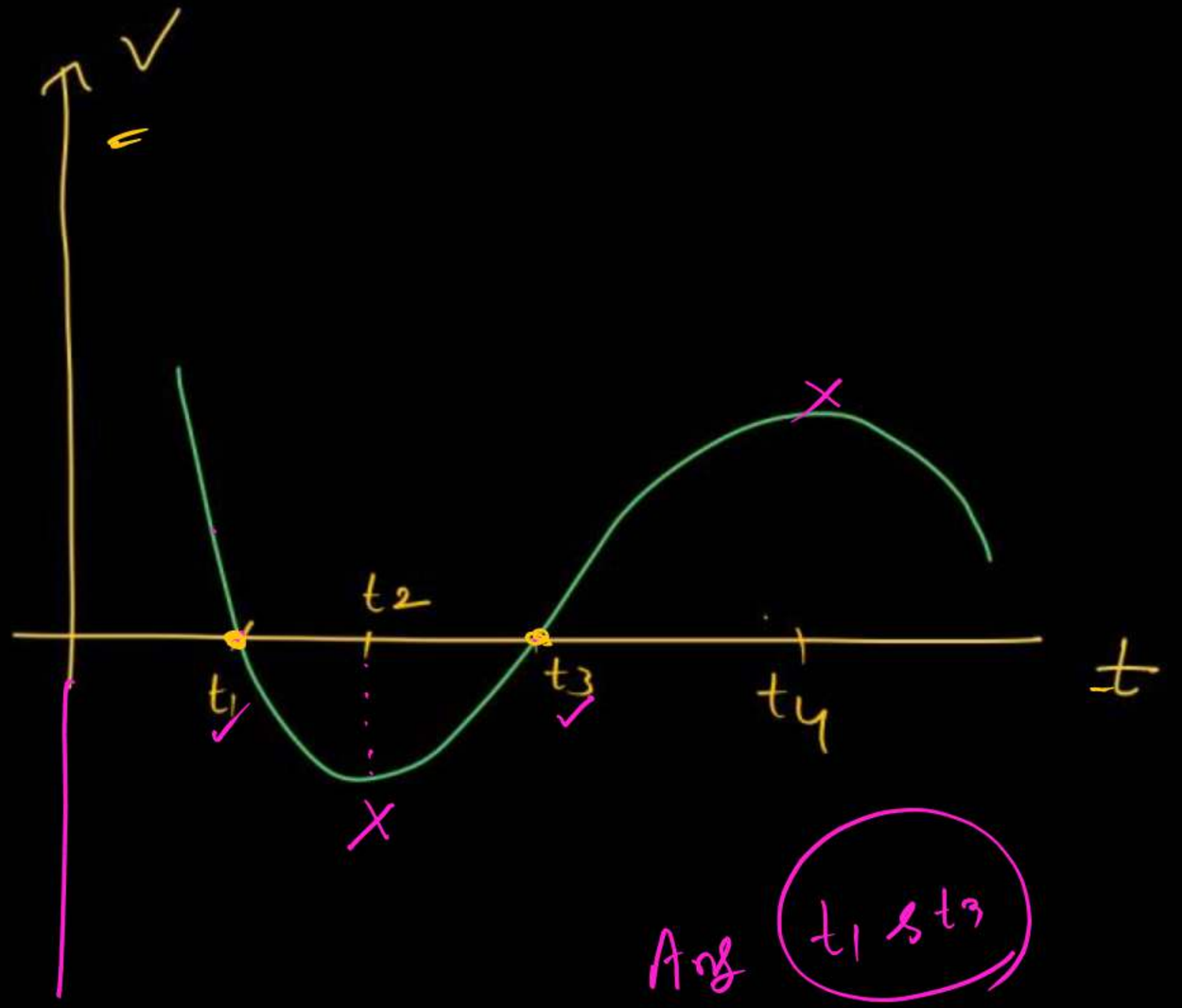
(i) $\boxed{V = 2\alpha(t) + \beta}$

$x(t=0) = 0 + 0 - \gamma = -\gamma m$

$V(t=0) = \beta$ m/s

(ii) $V_{t=2} = 2\alpha(2) + \beta = (4\alpha + \beta)$ m/s.

(42) object takes (takes) U-turn at



Ans t_1 & t_3

(43) Position of object $x = t^2 - 4t + 5$ then find distance and displacement in 3sec. Avg. speed & Avg velocity in 3-sec

Sol^n

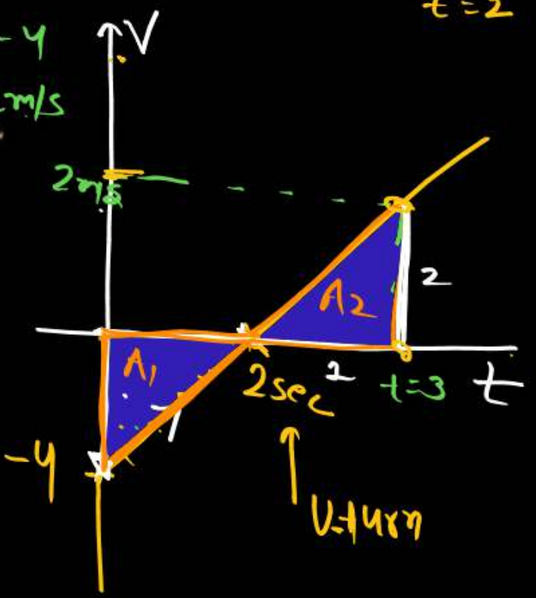
$$x = t^2 - 4t + 5$$

$$\hookrightarrow V = 2t - 4 \times 1 + 0$$

$$\boxed{V = 2t - 4} \quad y = mx + c$$

if $V = 0 = 2t - 4$, $t = 2$

$V_{t=3} = 6 - 4 = 2 m/s$



$A_1 = \frac{1}{2}(2)(-4) = -4$

$A_2 = \frac{1}{2} \times 1 \times 2 = 1m$

$dist^n = 5m$

$disp = -4 + 1 = -3m$

Avg speed $= \frac{5}{3} m/s$ ✓

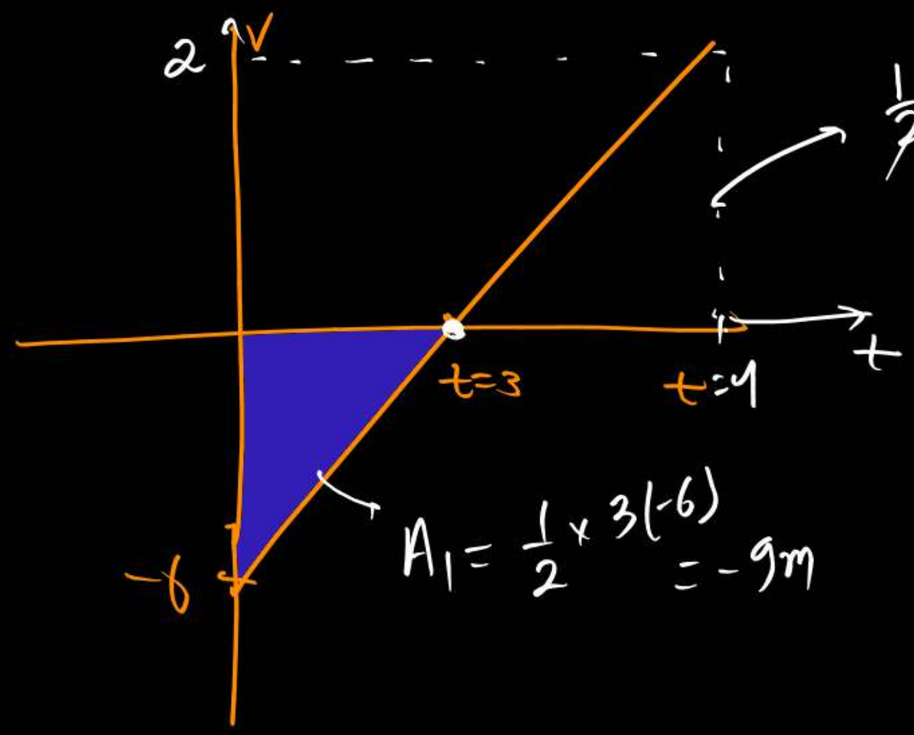
Avg velocity $= \frac{-3}{3} = -1 m/s$ ✓

(45) $x = t^2 - 6t + 4$ find Avg velocity in 4-sec.

→ Avg speed

Soln

$V = 2t - 6$ $\quad V_{t=4} = 2\times4 - 6 = 2$



$\frac{1}{2} \times 1 \times 2 = 1$

$A_1 = \frac{1}{2} \times 3(-6) = -9m$

dist^n $= 9 + 1 = 10m$ ✓

disp^n $= -9 + 1 = -8m$ ✓

Avg velocity $= \frac{-8m}{4} = -2m/s$

(46)

MR Scam

$x = \frac{t^3}{3} - 2t^2 + 4t$ find distance & displacement in 3-sec

$v = \frac{dx}{dt} = \frac{1}{3}(3t^2) - 2(2t) + 4$

$V = t^2 - 4t + 4 = (t-2)^2$

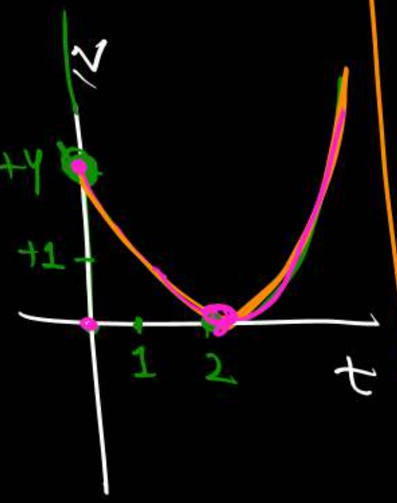
$0 = t^2 - 4t + 4$

$0 = t^2 - 2t - 2t + 4$

$0 = t(t-2) - 2(t-2)$

$0 = (t-2)(t-2)$

$t = 2$ Sec





$x(t=2) = \frac{8}{3} - 2(4) + 4 \times 2$

$= \frac{8}{3} - 8 + 8 = \frac{8}{3} = 2.66$

$x(t=3) = \frac{27}{3} - 2(3)^2 + 4 \times 3$

$= 9 - 2 \times 9 + 12$

$= 9 - 18 + 12 = -9 + 12$

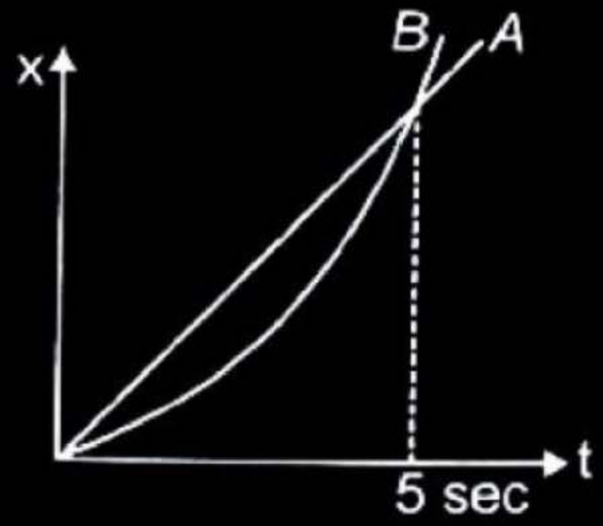
$= 3m$

Question

47

Position-time graph for two particles A and B as shown in figure. If both the particles start from origin, and average velocity of A and B during interval of 5 sec are v_A and v_B, then

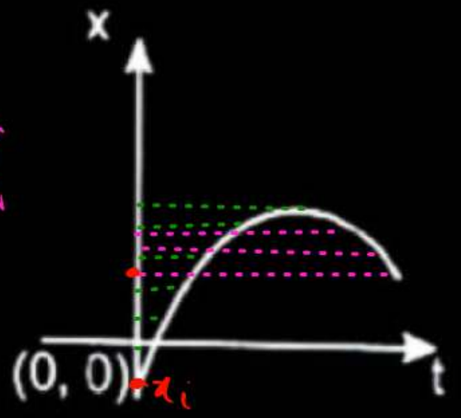
1. $v_A > v_B$
2. $v_B > v_A$
3. $v_A = v_B$
4. $v_A = 2v_B$



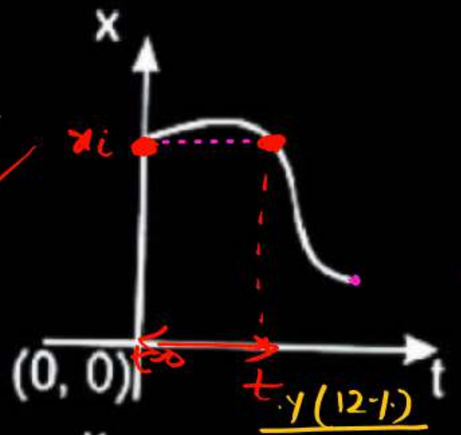
Question

Among the four graphs, there is only one graph for which average velocity = 0 is possible for a particular time interval starting from $t = 0$. Which one is it?

1



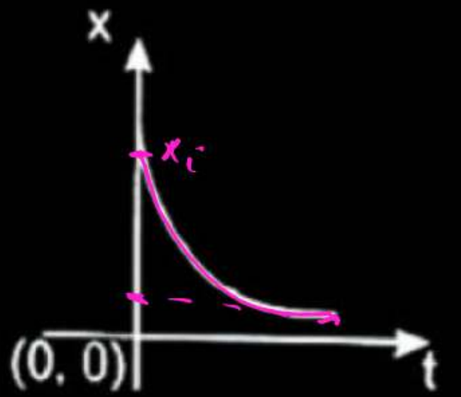
2



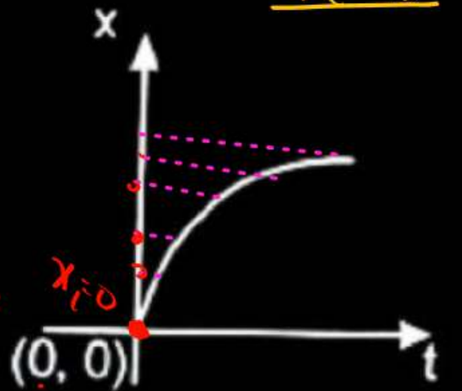
Avg velocity = $\frac{\text{Total displ}}{\text{(total time)}} = \frac{0}{\text{(total time)}} = 0$

$x_i = x_f$

3



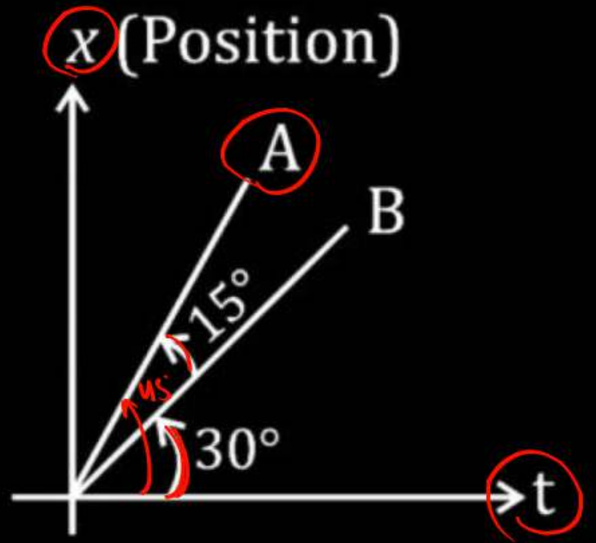
4



Question

45

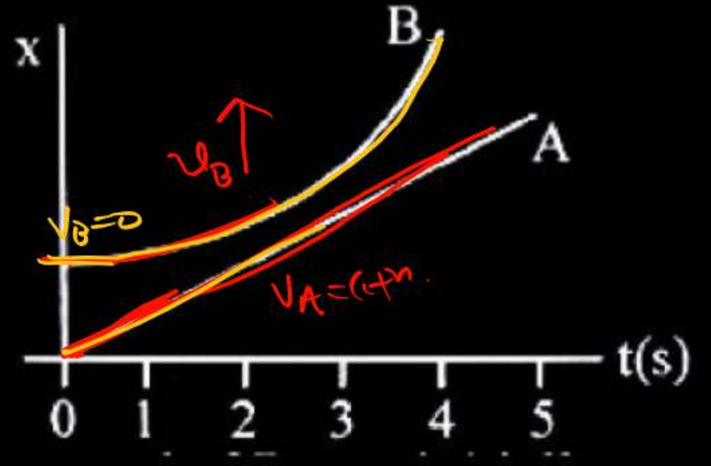
Find ratio of velocity of A to B.



$$\frac{V_A = \tan 45^\circ}{V_B = \tan 30^\circ}$$

Question

In the figure shown below, the position versus time graph of two particles A and B is shown. Select the correct statement:



1. The speed of B was initially greater than that of A and finally less than that of A.
2. The speed of B was initially less than that of A and finally greater than that of A.
3. The speed of B was initially as well as finally greater than that of A.
4. The speed of B was initially as well as finally less than that of A.

Question

A particle is moving such that its position coordinates (x, y) are (2m, 3m) at time $t = 0$, (6m, 7m) at time t = 2s and (13m, 14m) at time $t = 5$ s. The average velocity vector $\vec{v}_{avg}$ from t = 0 to t = 5 s is:

1. $\frac{1}{5}(13\hat{i} + 14\hat{j})$
2. $\frac{7}{3}(\hat{i} + \hat{j})$
3. $2(\hat{i} + \hat{j})$
4. $\frac{11}{5}(\hat{i} + \hat{j})$

Question

52.

Object is moving with constant speed then velocity of object:

1. may be variable
2. must be constant
3. must be variable
4. may be zero

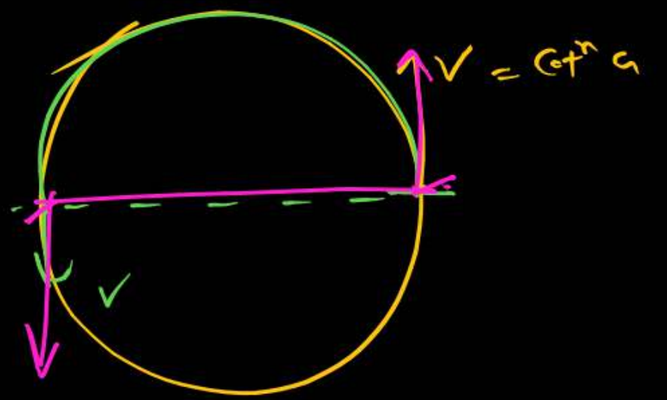
Question

53

A particle is executing a circular motion of radius R with a uniform speed v. After completing half the circle, the change in velocity and in speed will be respectively

vector

1. zero, zero (10/1) ✗
2. ✓ $2v$, zero
3. $2v$, $2v$ ✗
4. zero, $2v$ ✗



Question 54

If object is moving with speed $v = 3t^2$; then find avg. speed in 2-sec.

Instantaneous speed $V = 3t^2$

$$\langle v \rangle = \frac{\int v\,dt}{\int dt} = \frac{\int_0^2 3t^2\,dt}{\int_0^2 dt} = \frac{3\left[\frac{t^3}{3}\right]_0^2}{(t)_0^2} = \frac{(2)^3-(0)^3}{2-0}$$

$$= \frac{8}{2} = 4\ m/s$$

$$\# \quad \langle v \rangle = \frac{\int v\,dt}{\int dt}$$

(Q) Position of object $x = 5 + 2t^2$ (return) then find Avg. velocity b/w $t_1 = 2$ sec to $t_2 = 5$ sec.

Avg velocity $= \frac{\text{total disp}}{\text{total time}} = 14$ m/s ✓

not required method-2

Soln $V = \frac{dx}{dt} = 0 + 2(2t) = 4t$ ✓

$$\langle v \rangle = \frac{\int v\,dt}{\int dt} = \frac{4\int_2^5 t\,dt}{\int_2^5 dt} = \frac{4\left(\frac{t^2}{2}\right)_2^5}{(t)_2^5} = \frac{2\left[t^2\right]_2^5}{t} = 2(t)_2^5$$

wrong par scoring

$$= 2[5-2] = 2\times 3 = 6 \text{ m/s}$$

$$= \frac{2\left[t^2\right]_2^5}{(t)_2^5} = \frac{2\left[(5)^2-(2)^2\right]}{(5-2)} = \frac{2(25-4)}{3} = \frac{2}{3}\times 21^{7} = 14 \text{ m/s}$$

Motion ⇌ velocity

→ velocity ka feel hi motion ka feel hai

Acceleration = The rate of change in velocity w.r.t time = $\vec{a}_{inst}$
(motion)

$a = \frac{dv}{dt} \times \frac{dx}{dx} = \frac{dv}{dx} \times \frac{dx}{dt}$ ($\frac{dx}{dt} = v$)

→ ye motion ka feel nahi

→ change in velocity per sec ka feel.

$$\vec{a}_{inst} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{x}}{dt^2} = \vec{v}\left(\frac{d\vec{v}}{dx}\right)$$

$$\vec{a}_{Avg} = \frac{\vec{v}_f - \vec{v}_i}{\Delta t}$$

↳ vector, dirn of acc^n along change in velocity

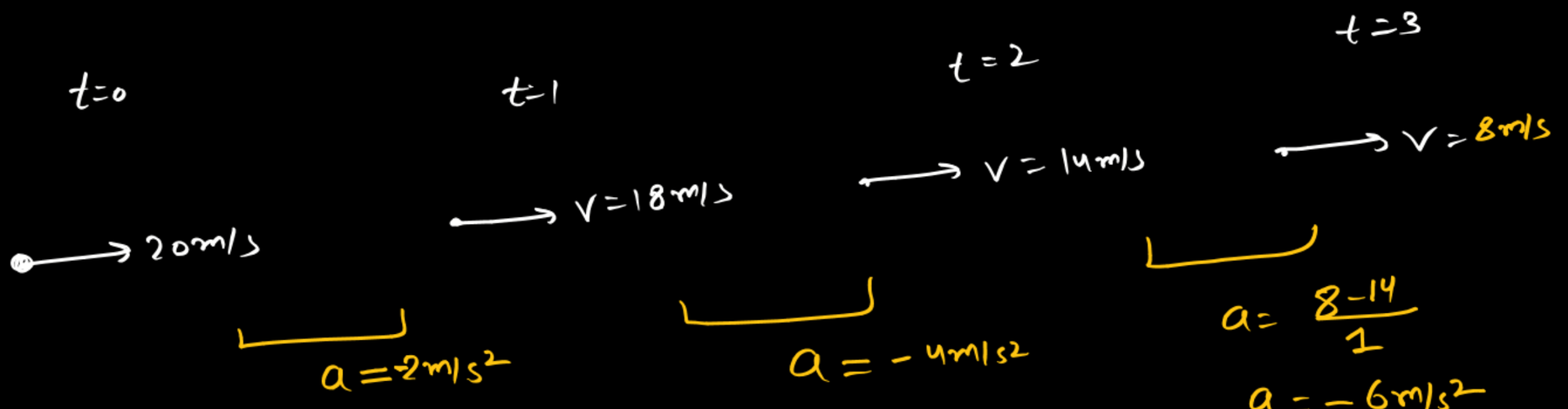
↳ along force $\vec{F} = m\vec{a}$

↳ m/s2

↳ $\langle a \rangle = \frac{\int a\,dt}{\int dt}$

copy

time $t=0$	$t=1s$	$t=2s$	$t=3s$	acceleration	
$\rightarrow$ 10m/s	$\rightarrow$ v=10m/s	$\rightarrow$ v=10m/s	$\rightarrow$ v=10m/s	$\vec{a}=\frac{\Delta v}{\Delta t}=0$	zero accn & Consin velocity (uniform motion)
$a=\frac{20-10}{1-0}=10m/s^2$ $\rightarrow$ 10m/s	$a=\frac{30-20}{2-1}=10m/s^2$ $\rightarrow$ v=20m/s	$a=\frac{40-30}{3-2}=10m/s^2$ $\rightarrow$ 30m/s	$\rightarrow$ 40m/s.	$\vec{a}=10m/s^2$ (uniform accn) Velocity ↑ (Increasing) ↳ Non uniform velocity	
$a=10m/s^2$ v=10m/s	$a=\frac{28-20}{1}=8m/s^2$ $\rightarrow$ 20m/s	$a=\frac{34-28}{3-2}=6m/s^2$ $\rightarrow$ v=28m/s	$\rightarrow$ v=34m/s	⇒ accn is continously decreasing (Non-uniform accn) ⇒ velocity is continously ↑ (Increasing)	
$a=10m/s^2$ $\rightarrow$ v=10m/s	$\rightarrow$ v=20m/s	$a=15m/s^2$ $\rightarrow$ v=35m/s	$a=20m/s^2$ $\rightarrow$ v=55m/s	accn is Velocity is increasing ↑	Continue ↑ (Both accn & velocity Non-uniform)
$\rightarrow$ v=20m/s	$\rightarrow$ v=10m/s	$\rightarrow$ v=6m/s	$\rightarrow$ v=4m/s	velocity (↓) accn (↓)	



#

velocity ↓

|acc| ↑

↳ magnitude

at any inst. $v = 0$ but $a \neq 0$

$v = 0$ → Motion

$a \neq 0$ → velocity me change

⇒ Possible at max^m height of motion under gravity

Copy

H/W

do in rough Page

Which of the following option is correct for velocity & acc^{n}

(a) $v \uparrow$; $a \downarrow$

(b) $v \uparrow$; $a = \text{cost}^{n}$

(c) $v \downarrow$; $a \downarrow$

(d) $v \uparrow$; $a \uparrow$

(e) $v \uparrow$; $a \uparrow$

(f) $v = \text{cost}^{n}$; $a \neq 0$

(g) $v \neq 0$; $a = 0$

(h) $v = 0$; $a \neq 0$

Question

Which of the following option is correct:

1. Velocity of object increasing and acceleration may decreasing.
2. Velocity of object decreasing and acceleration may increasing.
3. Acceleration may be non-zero when velocity of object is zero.
4. All of these.

Question

Object is moving such that its velocity and acceleration is in opposite direction then

1. Speed may constant.
2. Speed may increasing.
3. Speed must be decreasing.
4. Speed may be increasing or decreasing.

Question

An object is moving with constant velocity then which of the following option is correct

1. Acceleration may be increasing.
2. Acceleration is zero.
3. Acceleration is decreasing.
4. Acceleration is non-zero.

Question

Object is moving with acceleration 2 m/s^2 its velocity at $t = 0$ is 10 m/s then find its velocity at $t = 4$ sec.

Question

Velocity at t = 0 sec is 10 m/s its velocity becomes 40 m/s after 6 sec then find acceleration.

Question

Velocity at t = 2 sec is 20 m/s its t = 5 sec it becomes 32 m/s then velocity at 7 sec will be:

Question

Find acceleration in each term:

$x = 4t^2 + 6$	$x = 3t^2 + 4t + 6$	$x = 2t^3 + 5t$	$x = t^4 + 4t$
$v = 3t^2 + 4$	$v = 3t^2 + 4$	$v = t^3 + 4$	$v = 2x + 4$

(1) Position of object $x = t^3 - 6t^2 + 10$ find time when acceleration of object will be zero.

1. Object is moving with acceleration $2\ m/s^2$ its velocity at $t=0$ is $10\ m/s$ then find its velocity at $t = 4\ sec$.

2. Velocity at $t=0$ sec is $10\ m/s$ its velocity becomes $40\ m/s$ after 6 sec then find acceleration.

3. Velocity at $t = 2$ sec is $20\ m/s$ at $t = 5$ sec it becomes $32\ m/s$ then velocity at 7 sec. will be

Position of object $x = 2t^3 - 4t^2 + 4$

find velocity, acceleration and Position

at $t = 2$ Sec

gn Which case acceleration is uniform (constant)

(1) $x = \sqrt{t} + 4$

(2) $\sqrt{x} = t - 3$

(3) $x = t^{-2} + 4$

(4) $x = \frac{4}{t}$

(5) $x = t^3 + 4$

(6) $x = \frac{3}{t^2}$

(7) $x = 4t^2 + 6$

(8) $v = 3t^2 + 4$

(9) $v = 6t - 4$

(10) $v = \sqrt{x}$

(11) $v = x^2 + 4$

(12) $v = 2x + 3$

(13) $a = 2t$

(14) $a = 3x$

(15) $a = 2t^2$

(16) $a = 5 m/s^2$

(17) $a = 3\sqrt{x}$

(18) $x = A\sin(\omega t)$

(19) $v = A\cos(\omega t)$

(20) $a = 2\sin(t)$

Position $x = t^3 - 9t^2 + 4$. find velocity when acceleration is zero.

Position of object $x = 2t^3 - 24t + 4$ find acceleration when velocity of object is zero.


Thank
YOU